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Inventor(s)	Syuutou; Takahiro et al.

Blanking mold, blanking and stacking apparatus, and blanking method

Abstract

A die for a blanking device, a blanking accumulation device, and a blanking method are provided with which blanks having a length that is greater than a width of a metal coil can be blanked and accumulated in an accumulating device (piler), with a target orientation. The method includes a continuous metal coil that is received and is subjected to blanking using a die. The die has a shearing portion for blanking a blank. The shearing portion has an inclination angle relative to a line direction in which the blank is discharged. The die further includes, downstream of the shearing portion, a discharge means including an orientation changing means for changing an orientation of the blank from an inclined state having a first inclination angle, at the time at which shearing is complete, to a second larger inclination angle.

Inventors:	Syuutou; Takahiro (Saitama, JP), Nakashizu; Yoshitaka (Tochigi, JP)
Applicant:	G-TEKT CORPORATION (Saitama, JP)
Family ID:	1000008778832
Assignee:	G-TEKT CORPORATION (Saitama, JP)
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Primary Examiner: Nicholson, III; Leslie A

Attorney, Agent or Firm: Andrus Intellectual Property Law, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national stage application of International Application PCT/JP2023/015118, filed Apr. 14, 2023. The International Application claims priority of Japanese Patent Application No. 2022-109583, filed Jul. 7, 2022. The international application and Japanese application are both incorporated herein by reference, in entirety.

FIELD OF ART

(2) The present invention relates to a blanking mold, a blanking and stacking apparatus, and a blanking method.

BACKGROUND ART

(3) For the production of large-size molded products, such as front pillars, cross members, outer panels, and side door panels of vehicles, large size blanks are produced by a blanking apparatus at high speed, which are then fed to a press, for improved production efficiency.

(4) There is known an apparatus for conveying blanks, including a plurality of chain conveyors provided between a carry-in position into and carry-out position out of a mold of a press, for conveying blanks fed from a blanking apparatus, and lifting means provided on both sides of the mold, for lifting up and down the chain conveyors between a position higher than the position of the lower mold and a retracted position lower than the position of the lower mold (Patent Publication 1).

(5) This publication teaches to position a plate material for blanking onto a mold of a press by regulating the feed rates of the plurality of chain conveyors. Patent Publication 1: JP 3742804 B2

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(6) Blanks obtained by blanking a metal coil in a blanking (press) apparatus are tentatively stacked in a piler before being fed to a subsequent press.

(7) On the other hand, the workpieces produced by blanking conventionally had a dimension in the width direction of the metal coil the same as or smaller than the width of the metal coil.

(8) It is therefore a primary object of the present invention to provide a blanking mold, a blanking and stacking apparatus, and a blanking method, which realize blanking to produce blanks having a dimension in the width direction of the metal coil larger than the width of the metal coil, and stacking of the blanks in an intended orientation in the piler.

Means for Solving the Problems

(9) An aspect of the blanking apparatus which solves the above-mentioned problems is as follows.

(10) A blanking apparatus configured to perform blanking of a feed of metal coil continuous in a line direction, and to unload and convey resulting blanks, including: a mold including a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to the line direction in which the blanks are to be unloaded, and an unloader located downstream of the shearing section and having an orientation-adjuster for adjusting an orientation of the blanks from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, wherein the unloader having the orientation-adjuster includes a

first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor.

(11) Further, an aspect of the blanking and stacking apparatus is as follows.

(12) A blanking and stacking apparatus configured to perform blanking of a feed of continuous metal coil, and to unload and convey resulting blanks comprising: a mold including a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, an unloader located downstream of the shearing section and having an orientation-adjuster for adjusting an orientation of the blanks from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, a conveyor for the blanks located on an exit side of the unloader, and a piler for stacking in succession the blanks conveyed out of the conveyor into a pile, wherein the orientation-adjuster includes a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor.

(13) Further, an aspect of the first blanking method is as follows.

(14) A blanking method comprising: performing blanking of a feed of continuous metal coil in a blanking apparatus, and unloading and conveying resulting blanks, wherein the blanking apparatus includes a mold, and the mold includes: a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, and an orientation-adjuster located downstream of the shearing section in the line direction, for adjusting an orientation of the blanks on a conveying surface from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, and for guiding the blanks forward from the mold, wherein the orientation-adjuster includes a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor, and wherein the blanking is performed at an inclination angle selected from larger to smaller angles in proportion to a larger to smaller width of the metal coil to produce a predetermined length of the blanks.

(15) As aspect of the second blanking method is as follows. A blanking method including: performing blanking of a feed of continuous metal coil in a blanking apparatus, and unloading and conveying resulting blanks, wherein the blanking apparatus includes a mold, and the mold includes: a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, and an orientation-adjuster located downstream of the shearing section in the line direction for adjusting an orientation of the blanks on a conveying surface from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, and for guiding the blanks forward from the mold, wherein the orientation-adjuster includes a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor, and wherein the blanking is performed at an inclination angle selected from smaller to larger angles in proportion to a larger to smaller length of the blanks, with respect to a predetermined width of the metal coil.

Effect of the Invention

(16) According to the present invention, blanks having a dimension in the width direction of the

metal coil larger than the width of the metal coil may be produced through blanking, and stacked in the piler in an intended orientation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic plan view of an embodiment of the present invention.
- (2) FIG. 2 is a schematic plan view of the embodiment of the present invention for explaining the motion of the blanks.
- (3) FIG. 2A is an explanatory view of an installable area for the unloader having an orientation-adjuster.
- (4) FIG. 2B is an other explanatory view of an installable area for the unloader having an orientation-adjuster.
- (5) FIG. 3 is a schematic front view of the shearing section.
- (6) FIGS. 4(a) to 4(c) illustrate in plan views embodiments of the blanking in view of the width of the metal coil.
- (7) FIG. 5 illustrates conveying and stacking of the blanks, wherein FIG. 5(a) is a plan view and FIG. 5(b) is a schematic front view.
- (8) FIG. 6 illustrates conveying and stacking of the blanks without adjusting their orientations, wherein FIG. 6(a) is a plan view and FIG. 6(b) is a schematic front view.
- (9) FIG. 7 is a schematic plan view of an embodiment of the unloader having an orientation-adjuster.
- (10) FIG. 8 is a schematic plan view of an embodiment of the unloader having an alternative adjuster.
- (11) FIG. 9 is a schematic front view of an embodiment of the unloader having an orientation-adjuster.
- (12) FIG. 10 is a schematic plan view of an embodiment of the unloader having an alternative orientation-adjuster.
- (13) FIG. 11 is a schematic plan view of an embodiment of the unloader having a still alternative orientation-adjuster.
- (14) FIG. 12 is an explanatory view of a first embodiment wherein the blanking is performed at an inclination angle selected in proportion to the length of the blanks to be produced, with respect to the predetermined width of metal coil.
- (15) FIG. 13 is an explanatory view of a second embodiment wherein the blanking is performed at an inclination angle selected in proportion to the length of the blanks to be produced, with respect to the predetermined width of metal coil.

MODE FOR CARRYING OUT THE INVENTION

- (16) Embodiments of the present invention will now be discussed with reference to the attached drawings.
- (17) FIG. 1 shows a blanking apparatus 1, with its upper mold being omitted and only its lower mold 20 being shown in a plan view.
- (18) The blanking apparatus 1 is configured to perform blanking of a feed of continuous metal coil 10, and to unload and convey the blanks 11 from the mold 20.
- (19) Referring to FIG. 2, the mold 20 has a shearing section 12, wherein blanking the metal coil 10 to produce blanks 11 is performed at an inclination angle $\theta 1$ with respect to the line direction CL in which the blanks 11 are to be unloaded. The shearing section 12 includes, as shown in FIG. 3, a lower blade (blanking die) 12A of the lower mold and an upper blade (blanking punch) 12C of the upper mold. Referred to by the reference numeral 10S is scrap.
- (20) The mold 20 also has, downstream of the shearing section 12, unloader 30 having an

orientation-adjuster for adjusting the orientation of the blanks **11** from an inclined state upon completion of the shearing at a first inclination angle θ_1 with respect to the line direction CL to a larger, second inclination angle θ_2 which allows stacking of the blanks **11** in alignment with each other in a generally rectangular piler **41** as shown in FIG. 2, preferably perpendicular to the line direction CL.

(21) Here, angle θ_2 minus angle θ_1 may preferably be less than 90 degrees and 30 degrees or more.

(22) Also provided is a conveyor **40** for the blanks **11** located on the exit side of the unloader **30**, and a piler **41** for stacking in succession the blanks **11** conveyed out of the conveyor **40** into a pile.

(23) The mold **10** according to the embodiment is a progressive mold, which has shearing sections **12** in line with the shearing steps along the feeding direction of the metal coil **10**.

(24) For example, in order to form an inclined, indented contour, a first step of punching is performed by shearing part of the indented contour to form punched areas **11a1** to **11a3**, and subsequently a second step of contour machining is performed for forming contour-machined areas **11b1** to **11b3** to completely shear the contour to obtain individual blanks **11**.

(25) In forming blanks with a simple contour, the shearing process may be performed with one blade without employing a progressive mold.

(26) FIGS. 1 and 2 have been discussed taking the blanks for a cross member as an example. Now referring to FIG. 4, layouts of blanks are discussed taking blanks for a front pillar as an example.

(27) When the length L of the blanks B is shorter than the width W₀ of the metal coil **10**, the blanking may be performed successively, with the blanks being aligned to the width direction of the metal coil **10** as in the layout examples shown in FIGS. 4(a) and 4(b) for utilizing the overall metal coil **10** as much as possible to improve the yield.

(28) On the other hand, in the case of blanks B which are longer than the width W of the metal coil **10** or which leave little bridges on both sides in the direction of the width W, the blanks B should be arranged in an inclined layout, for example, as shown in FIG. 4(c). Alternatively, it may be envisaged to use a wider metal coil **10** in proportion to the length L of the blanks B. However, in view of limited variation of width available for metal coils **10**, it is desirable to layout the blanks B in an inclined fashion, for example, as shown in FIG. 4(c), for improving the yield.

(29) Here, when the blanking is performed with an inclined layout of the blanks B, for example, as shown in FIG. 4(c), the blanks obtained from the blanking are unloaded from the mold, conveyed on a conveyor **40**, such as a belt conveyor, and stacked in a piler **41**. Meanwhile, on the conveyor **40**, the blanks B are conveyed in the orientation in which they were first brought into contact with the top surface of the conveyor **40**, such as a belt conveyor. Moreover, large-size blanks B are conveyed at an angle inclined with respect to the line direction, so that individual blanks B are conveyed to the piler without being aligned to each other in the conveying direction, and allowed to fall therein to be stacked into a pile.

(30) The individual blanks B, when allowed to fall in the piler **41** to be stacked into a pile without being aligned to each other in the conveying direction, may not be (properly) stacked in an intended orientation in the piler. In addition, as the blanks are inclined, the piler needs to be larger in size.

(31) Consequently, it is necessary to stack the blanks in an intended orientation successively in order. According to an embodiment of the invention, based on the idea of adjusting (correcting) the orientation of the blanks, orientation-adjusting and unloader **30** is disposed in the mold **20**.

(32) Here, it is conceivable to adjust (correct) the orientation of the blanks after the blanks are unloaded from the mold **20**. In other words, it is conceivable to dispose the blank-orientation-adjuster (corrector) between the mold **20** and the piler **41**.

(33) However, disposing the blank-orientation-adjuster (corrector) between the mold **20** and the piler **41** would require drastic change of the system, and should thus be avoided. Further, the blanks that have gone through the final shearing step are loose, and their orientation may not become stable before being unloaded from the mold **20**.

(34) It is also probable that the blanks **B** may fall into the “installable area” and no longer be conveyed.

(35) On the other hand, disposing the orientation-adjusting and unloader **30** within the mold **20** allows stable adjustment of the orientation as well as stable unloading and conveying, and does not require drastic change of the system, which leads to great economical advantage. In addition, in response to the alteration of the blanking products, the operation control of the orientation-adjusting and unloader **30** may simply be modified, or the mold may simply be replaced with a mold **20** equipped with the orientation-adjusting and unloader **30**, which is highly desirable in view of production efficiency. Alternatively, an existing mold may be provided with the orientation-adjusting and unloader **30**, utilizing an empty space, i.e., the “installable area” as will be discussed later. This is particularly advantageous for continuous production of inclined blanks in a compact space.

(36) Next, the substance of the present invention will now be discussed with reference to some specific embodiments.

First Embodiment

(37) FIGS. **1** to **3** show a first embodiment, whose structure is as discussed above. This first embodiment allows high-speed production by performing the blanking process, adjustment of orientation, and stacking, with the blanks being arranged in the inclined layout as in FIG. **4(c)** for improved yield.

(38) The mold **20** is an example of a progressive mold, and has a shearing section **12** in which the blanking is performed to produce the blanks **11** at an inclination angle of θ_1 with respect to the line direction CL, in which a metal coil **10** is fed from an uncoiler and the blanks **11** are to be unloaded. The details of the upper mold is not shown, and a commonly-known structure may be employed.

(39) The mold **20** has, downstream of the shearing section **12**, unloader **30** having an orientation-adjuster for adjusting the orientation of blanks **11** from an inclined state upon completion of the shearing at a first inclination angle θ_1 with respect to the line direction CL to a larger, second inclination angle θ_2 which allows stacking of the blanks **11** in alignment with each other in rectangular piler **41** as shown in FIG. **2**, preferably perpendicular to the line direction CL.

(40) In the first embodiment, the unloader **30** having the orientation-adjuster has a first belt conveyor **31** with a longer conveying distance and a second belt conveyor **32** with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks **11** are conveyed, with the conveying rate of the first belt conveyor **31** being faster than the conveying rate of the second belt conveyor **32**.

(41) Here, the conveying rate of the first belt conveyor **31** in the line direction and the conveying rate of the second belt conveyor **32** in the line direction are the same as or higher than the conveying rate of the metal coil in the line direction, so that the blanks may be unloaded without accumulating.

(42) The unloader **30** having the orientation-adjuster of the first embodiment has both the functions to adjust the orientation of and to unload and convey the blanks.

(43) That is, during the transfer by means of the first belt conveyor **31** with a longer conveying distance and the second belt conveyor **32** with a shorter conveying distance, with the conveying rate of the first belt conveyor **31** being faster than the conveying rate of the second belt conveyor **32**, the respective blanks **11** may be rotated counterclockwise as seen in FIG. **2**, while the blanks **11** are unloaded from the mold **20** into the rectangular piler **41** in alignment with each other.

(44) The unloader **30** having the orientation-adjuster may have the orientation-adjuster and the unloader separately composed. The orientation-adjuster may be composed of a turn table, while the unloader may be composed of the belt conveyor or a chute.

(45) The blanks **11** unloaded from the mold **20** are conveyed by means of the conveyor **40** for blanks **11**, such as a belt conveyor or a chute, located at the exit side of the unloader **30**, and successively stacked into a pile in the piler **41**.

(46) As shown in FIGS. 2 and 3, by positioning the entry side of the blank-conveyor **40** below and underlapping the exit side of the unloader **30**, the blanks **11** whose orientations have been adjusted may be allowed to fall onto the conveyor **40** for high-speed unloading and conveying.

(47) The piler **41** is provided, for example as in the embodiment shown in FIG. 5, with plate-like positioning stampers **41a**, of which cylinders are driven to lift up and down the stampers **41a**, at each longitudinal end of the blanks B of which orientations have been adjusted. The stampers **41a** are disposed on the surfaces along the thickness direction of the blanks B, two facing each other at each longitudinal end and positioning the blanks B at least from two directions on each of the facing side. Such arrangement allows easy regulation of the positioning.

(48) On the other hand, as in the embodiment shown in FIG. 6, if the positioning is to be regulated at four locations (referred to by reference sign X) by stampers **42a**, **42a** of the shape as shown in FIG. 6 for stacking the blanks in piler **42**, for example, without adjusting the orientations of the blanks B (as they are at the first inclination angle $\theta 1$), the blanks B being conveyed by the conveyor **40** may fall forward and be bogged down short of the pile, and thus may not be stacked in the right orientation. In addition, the piler **42** wastes space to deteriorate space efficiency at the production site.

(49) Therefore, the adjustment of the orientation is important.

Second Embodiment

(50) The unloader **30** having the orientation-adjuster is disposed in “installable area Z” of the mold **20**.

(51) It suffices that the unloader **30** having the orientation-adjuster is disposed in its major part with respect to the mold **20** or the “installable area Z” and, for example, as seen from above, the unloader **30** having the orientation-adjuster may extend out of the mold **20**.

(52) Here, an example of the “installable area Z” is discussed.

(53) A first example of the installable area is an area, as shown in FIG. 2A, bounded by the edge **11e** of a blank **11** (or B) on its unloading side at an inclination angle $\theta 1$ (see FIG. 2), the front end face **20a** of the mold **20** on its unloading side, and side faces **20b**, **20c** of the mold **20**.

(54) Seen schematically, the edge **11e** of a blank **11** (or B) on its unloading side corresponds to the line A-B, the front end face **20a** of the mold **20** on its unloading side corresponds to the line C-D, and the side face **20b** of the mold **20** corresponds to the line D-B while the side face **20c** corresponds to the line A-C, so that the trapezoidal area bounded by the lines A-B-D-C is an installable area Z1.

(55) A second example of the installable area is an area, as shown in FIG. 2B, bounded by the edge **11e** of a blank **11** (or B) on its unloading side at an inclination angle $\theta 1$, which is smaller than that in FIG. 2A, the front end face **20a** of the mold **20** on its unloading side, and the side face **20b** of the mold **20**.

(56) Seen schematically, the edge **11e** of a blank **11** (or B) on its unloading side corresponds to the line A-B, the front end face **20a** of the mold **20** on its unloading side corresponds to the line A-D, and the side face **20b** of the mold **20** corresponds to the line D-B, so that the triangle area bounded by the lines A-B-D is an installable area Z2.

(57) As contrastively illustrated in the first and second examples, the installable area is formed corresponding to the shearing of blanks **11** (or B) at an inclination angle, and this area varies depending on the first inclination angle $\theta 1$ of blanks **11** (or B), the position of the blanks **11** (or B) with respect to the line direction CL, the profile of the edge **11e** of the blanks **11** (or B) on its unloading side, or the like factors.

(58) However, the installable area may be said to be basically a triangle area.

(59) As discussed regarding the second embodiment, by disposing the unloader **20** having the orientation-adjuster in the installable area X, which was originally an idle space in the mold **20** other than the shearing section **12**, an area having an inclination angle in the mold is effectively used, and both the adjustment of the orientations of the blanks and the unloading of the blanks in an

orientation aligned to each other into the piler **41** are achieved at the same time. As a result, the blanks may be stably stacked in the piler **41**.

Third Embodiment

(60) The unloader **30** having the orientation-adjuster is equipped with, as shown in FIG. 7, three or more belt conveyors **31**, **32**, **33** arranged substantially in parallel in the order of the conveying distance for which the blanks are conveyed on the conveyors, wherein the conveying rate of a belt conveyor is faster with a longer conveying distance of the belt conveyor and slower with a shorter conveying distance, i.e., $31.fwdarw.32.fwdarw.33$.

(61) Further, the belt conveyors in the order of **31**, **32**, **33** may be arranged along the line direction CL as shown in the figure (the same may be applied also to other embodiments).

(62) According to the third embodiment, the blanks, which are supported by three or more belt conveyors **31**, **32**, **33**, undergo the orientation adjustment with little warping, so that the orientation adjustment may be stable.

Fourth Embodiment

(63) As shown in FIG. 8, the unloader **30** having the orientation-adjuster may have a first belt conveyor **31** with a longer conveying distance and a second belt conveyor **32** with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, and the first belt conveyor **31** may have at least two split conveyors **31A**, **31B** divided in the conveying direction, and the split conveyors **31A** and **31B** may be different in conveying rate.

(64) According to this fourth embodiment, through setting or changing the speeds of the split conveyor **31A**, the split conveyor **31B**, and the second belt conveyor **32**, the adjustment of the orientation may be controlled precisely.

Fifth Embodiment

(65) The height of the belt conveyors **31**, **32** may be different. For example, as shown in FIG. 9, the first belt conveyor **31** may be positioned at a higher level than that of the second belt conveyor **32**.

(66) In such a configuration, during swiveling of blanks for adjusting their orientation, more weight is applied to the second belt conveyor **32** with a shorter conveying distance while less weight is applied to the first belt conveyor **31** with a longer conveying distance, so that adjustment of the orientation involving swiveling or correction of warping of the blanks may be performed easily.

(67) The latter, correction of warping of the blanks is discussed in detail. Blanks may be warped in one direction depending on the layout of the punching blades. In this case, by positioning the belt conveyor located under the side of the blank which would be warped and sagged as it stands, for example, the first belt conveyor **31**, at a higher level than the level of the second belt conveyor **32**, the blanks may quickly be mounted on a conveyor (the first belt conveyor **31** in this embodiment), which allows correction of the warping.

Sixth Embodiment

(68) As shown in FIG. 10, the unloader **30** having the orientation-adjuster may have a first belt conveyor **31** with a longer conveying distance and a second belt conveyor **32** with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, each blank B with respect to its longitudinal direction has a first portion B1 having a larger area of contact with a conveyor and a second portion B2 having a smaller area of contact with a conveyor, and the first portion B1 is associated with the second belt conveyor **32**, whereas the second portion B2 is associated with the first belt conveyor **31**.

(69) Even the speed of the second belt conveyor **32** is increased, stable conveyance is performed, as the area of contact between the first portion B1 and the second belt conveyor **32** is larger. Further, as the conveying distance of the first belt conveyor **31** is longer, even the centrifugal force generated by the swiveling acts on the blanks, displacement or slipping of the blanks may be regulated to allow stable adjustment of the orientation.

Seventh Embodiment

(70) As shown in FIG. 11, the unloader **30** having the orientation-adjuster may have a first belt

conveyor **31** with a longer conveying distance and a second belt conveyor **32** with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, each blank **B** with respect to its longitudinal direction has a first portion **B1** having a larger area of contact with a conveyor and a second portion **B2** having a smaller area of contact with a conveyor, and the first portion **B1** is associated with the first belt conveyor **31**, whereas the second portion **B2** is associated with the second belt conveyor **32**.

(71) As the first portion **B1** having a larger area of contact, which is also heavier, is carried on the first belt conveyor **31** with a longer conveying distance, even the centrifugal force generated by the swiveling acts on the blanks, displacement thereof may be regulated due to the friction force to allow stable adjustment of the orientation.

(72) Here, selection between the sixth and seventh embodiments may be made depending on the shape of the blanks or the like.

(73) Note that the blanks may have a laterally symmetric shape.

Eighth Embodiment

(74) The above-discussed first to eighth embodiments may suitably be incorporated into the blanking and stacking apparatus according to the present invention.

(75) Referring to FIGS. **1** to **3** illustrating a typical embodiment, there is provided a blanking apparatus **1** having a mold and configured to perform blanking of a continuous metal coil **10** fed from an uncoiler (not shown), and to unload and convey the resulting blanks **11** (**B**) from the mold **20**.

(76) The mold **20** has a shearing section **12**, wherein blanking to produce blanks **11** is performed at an inclination angle $\theta 1$ with respect to the line direction **CL** in which the blanks **11** (**B**) are to be unloaded. The shearing section **12** includes, for example, a lower blade (blanking die) **12A** of the lower mold and an upper blade (blanking punch) **12B** of the upper mold, as shown in FIG. **3**.

(77) The mold **20** has, downstream of the shearing section **12**, unloader **30** having an orientation-adjuster for adjusting the orientation of blanks **11** from an inclined state upon completion of the shearing at a first inclination angle $\theta 1$ with respect to the line direction **CL** to a larger, second inclination angle $\theta 2$, preferably perpendicular to the line direction **CL**.

(78) Further provided on the exit side of the unloader is a conveyor **40** for blanks **11**(**B**), and a piler **41** for stacking in succession the blanks **11** (**B**) conveyed out of the conveyor **40** into a pile.

(79) As shown in FIGS. **2** and **3**, the entry side of the blank-conveyor **40** is positioned below and underlapping the exit side of the unloader **30**, so that the blanks **11** whose orientations have been adjusted may be allowed to fall onto the conveyor **40** for high-speed unloading and conveying.

Ninth Embodiment

(80) According to the present invention, there is provided a blanking method, for example, as follows.

(81) That is, the blanking method includes performing blanking of a feed of continuous metal coil **10** in a blanking apparatus **1** having a mold **20**, and unloading and conveying the resulting blanks **11** (**B**) from the mold **20**, and the mold **20** of the blanking apparatus **1** includes, downstream of the shearing section **12**, unloader **30** having an orientation-adjuster for adjusting the orientation of blanks **11** from an inclined state upon completion of shearing at a first inclination angle $\theta 1$ with respect to the line direction **CL** to a larger, second inclination angle $\theta 2$, preferably perpendicular to the line direction **CL**.

(82) The unloader **30** having the orientation-adjuster is disposed in the “installable area **Z**” of the mold **20** as discussed above.

(83) According to this embodiment, with respect to the predetermined width of metal coil **10**, the inclination angle is selected for the blanking process, in proportion to the length of the product to be produced by the blanking.

(84) In this case, as shown in FIG. **12**, the length and the inclination angle of the blanks **B** may be selected depending on the variation of width of the metal coil **10** for making the most of the area of

the metal coil **10**.

(85) On the other hand, as shown in FIG. **13**, when the predetermined width of the metal coil **10** is limited, the inclination angle of the blanks B may be selected for making the most of the area of the metal coil **10**.

(86) In this way, by means of an existing apparatus, longer blanks B than those which have been produced in the apparatus, or longer blanks B than the constant width of the metal coil being fed, may be produced, which enhances flexibility in designing and production.

DESCRIPTION OF REFERENCE NUMERALS

(87) **1**: blanking apparatus **10**: metal coil **11**: blank **11a1-11a3**: punched area **11b1-11b3**: contour-machined area **12**: shearing section **12A**: lower blade (blanking die) **12B**: upper blade (blanking punch) **20**: mold **30**: unloader having orientation-adjuster **40**: conveyor **41**: piler $\theta 1$: first inclination angle $\theta 2$: second inclination angle CL: line direction Z, Z1, Z2: installable area

Claims

1. A blanking apparatus configured to perform blanking of a feed of metal coil continuous in a line direction, and to unload and convey resulting blanks, comprising: a mold comprising a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to the line direction in which the blanks are to be unloaded, and an unloader located downstream of the shearing section and having an orientation-adjuster for adjusting an orientation of the blanks from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, wherein the unloader having the orientation-adjuster comprises a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor.
2. The blanking apparatus according to claim 1, wherein the unloader having the orientation-adjuster is disposed in its major part in an installable area bounded by an edge of a blank on its unloading side at the first inclination angle, a front end face of the mold on its unloading side, and side faces of the mold.
3. The blanking apparatus according to claim 2, wherein the first belt conveyor is arranged closer to a longer of the side faces defining the installable area, and the second belt conveyor is arranged closer to a shorter of the side faces defining the installable area.
4. The blanking apparatus according to claim 1, wherein the conveying rate of the second belt conveyor in the line direction is same as or higher than a feeding rate of the metal coil in the line direction.
5. The blanking apparatus according to claim 1, wherein the orientation-adjuster comprises three or more belt conveyors arranged substantially in parallel in an order of conveying distance for which the blanks are conveyed, and wherein a conveying rate of a belt conveyor is faster in an order of shorter to longer conveying distance of the belt conveyor.
6. The blanking apparatus according to claim 1, wherein the orientation-adjuster comprises the first belt conveyor with the longer conveying distance and the second belt conveyor with the shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, and wherein the first belt conveyor comprises at least two split conveyors divided in a conveying direction, and the split conveyors are different in conveying rate.
7. The blanking apparatus according to claim 1, wherein one of the first belt conveyor and the second belt conveyor is positioned at a higher level than a level of the other.
8. The blanking apparatus according to claim 1, wherein the orientation-adjuster comprises the first belt conveyor with the longer conveying distance and the second belt conveyor with the shorter conveying distance, which are arranged substantially in parallel and on which the blanks are

conveyed, wherein each blank with respect to its longitudinal direction has a first portion having a larger area of contact with a conveyor and a second portion having a smaller area of contact with a conveyor, and wherein the first portion is associated with the second belt conveyor, whereas the second portion is associated with the first belt conveyor.

9. The blanking apparatus according to claim 1, wherein the orientation-adjuster comprises the first belt conveyor with the longer conveying distance and the second belt conveyor with the shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, wherein each blank with respect to its longitudinal direction has a first portion having a larger area of contact with a conveyor and a second portion having a smaller area of contact with a conveyor, and wherein the first portion is associated with the first belt conveyor whereas the second portion is associated with the second belt conveyor.

10. A blanking and stacking apparatus comprising: a blanking apparatus comprising a mold and configured to perform blanking of a feed of continuous metal coil, and to unload and convey resulting blanks from the mold, wherein the mold comprises a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, and an unloader located downstream of the shearing section and having an orientation-adjuster for adjusting an orientation of the blanks from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, an conveyor for the blanks located on an exit side of the unloader, and a piler for stacking in succession the blanks conveyed out of the conveyor into a pile, wherein the orientation-adjuster comprises a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor.

11. The blanking and stacking apparatus according to claim 10, wherein the unloader having the orientation-adjuster is disposed in its major part in an installable area bounded by an edge of a blank on its unloading side at the first inclination angle, a front end face of the mold on its unloading side, and side faces of the mold.

12. A blanking method comprising: performing blanking of a feed of continuous metal coil in a blanking apparatus, and unloading and conveying resulting blanks, wherein the blanking apparatus comprises a mold, and the mold comprises: a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, and an orientation-adjuster located downstream of shearing section in the line direction, for adjusting an orientation of the blanks on a conveying surface from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, and for guiding the blanks forward from the mold, wherein the orientation-adjuster comprises a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor, and wherein the blanking is performed at an inclination angle selected from larger to smaller angles in proportion to a larger to smaller width of the metal coil to produce a predetermined length of the blanks.

13. A blanking method comprising: performing blanking of a feed of continuous metal coil in a blanking apparatus, and unloading and conveying resulting blanks, wherein the blanking apparatus comprises a mold, and the mold comprises: a shearing section for performing blanking of the feed of metal coil to produce blanks at an inclination angle with respect to a line direction in which the blanks are to be unloaded, and an orientation-adjuster located downstream of the shearing section in the line direction for adjusting an orientation of the blanks on a conveying surface from an inclined state at a first inclination angle with respect to the line direction to a larger, second inclination angle, and for guiding the blanks forward from the mold, wherein the orientation-

adjuster comprises a first belt conveyor with a longer conveying distance and a second belt conveyor with a shorter conveying distance, which are arranged substantially in parallel and on which the blanks are conveyed, with a conveying rate of the first belt conveyor being faster than a conveying rate of the second belt conveyor, and wherein the blanking is performed at an inclination angle selected from smaller to larger angles in proportion to a larger to smaller length of the blanks, with respect to a predetermined width of the metal coil.
